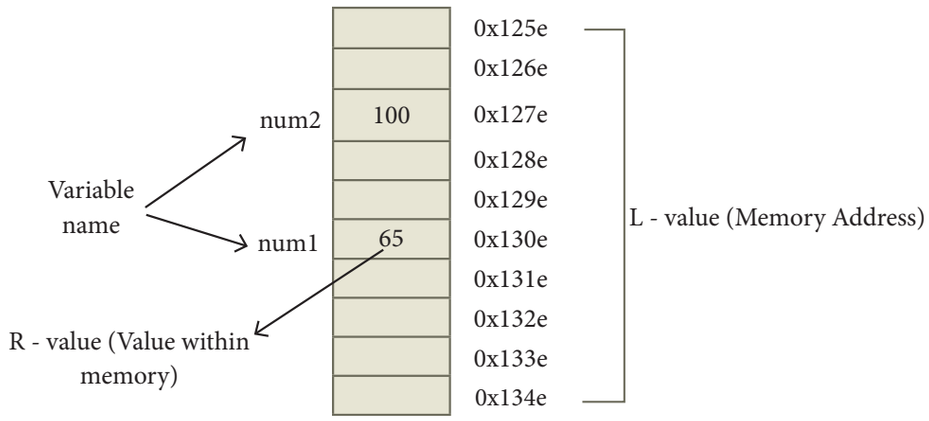

Figure 9.14 Memory allocation of a variable

Remember that, the memory addresses are in the form of Hexadecimal values

9.13.1 Declaration of Variables

Every variable should be declared before they are actually used in a program. Declaration is a process to instruct the compiler to allocate memory as per the type that is specified along with the variable name. For example, if you declare a variable as int type, in Dev C++, the compiler allocates 4 bytes of memory. Thus, every variable should be declared along with the type of value to be stored.

Declaration of more than one variable:

More than one variable of the same type can be declared as a single statement using a comma separating the individual variables.

Syntax:

<data type> <var1>, <var2>, <var3> …… <var_n>;

Example:

int num1, num2, sum;

In the above statement, there are three variables declared as int type. Which means, in **num1, num2** and **sum**, you can store only integer values.

For the above declaration, the C++ compiler allocates 4 bytes of memory (i.e. 4 memory boxes) for each variable.

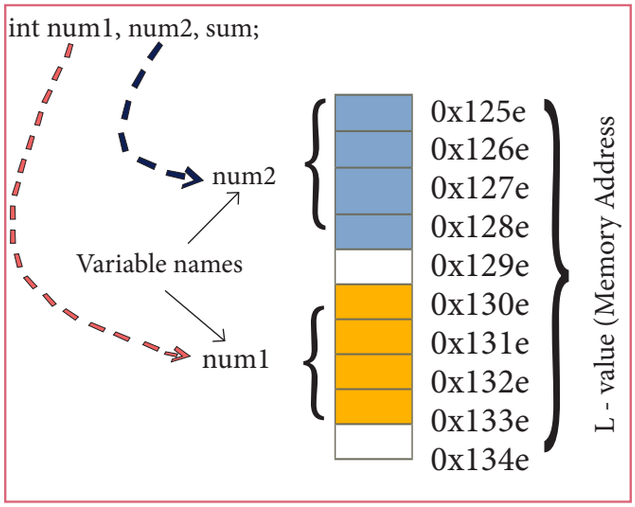

Figure 9.15 Memory allocation of int type variables

Chapter 9 Page 115-151.indd 139 3/24/2020 9:21:05 AM